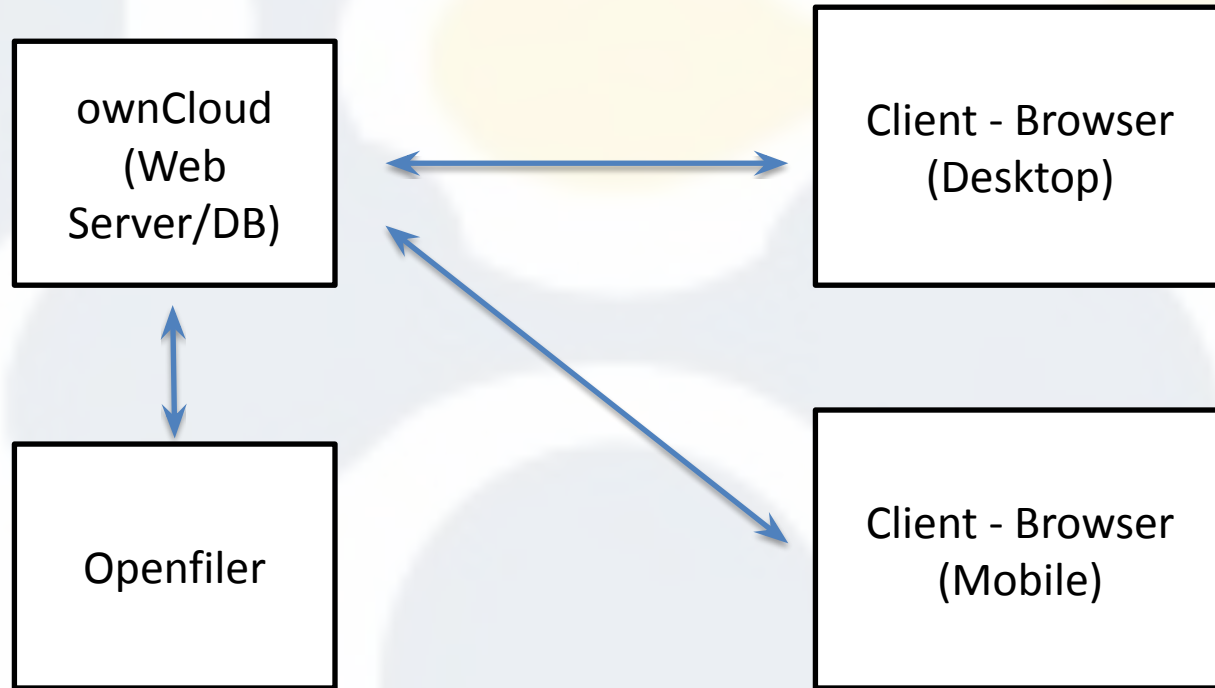


NAS dengan Openfiler dan ownCloud

Topologi



Instalasi Openfiler

Booting via Netboot PENS

- Misc Openfiler (password: nas)

```
...:: EEPIS Netboot Computer System :::...

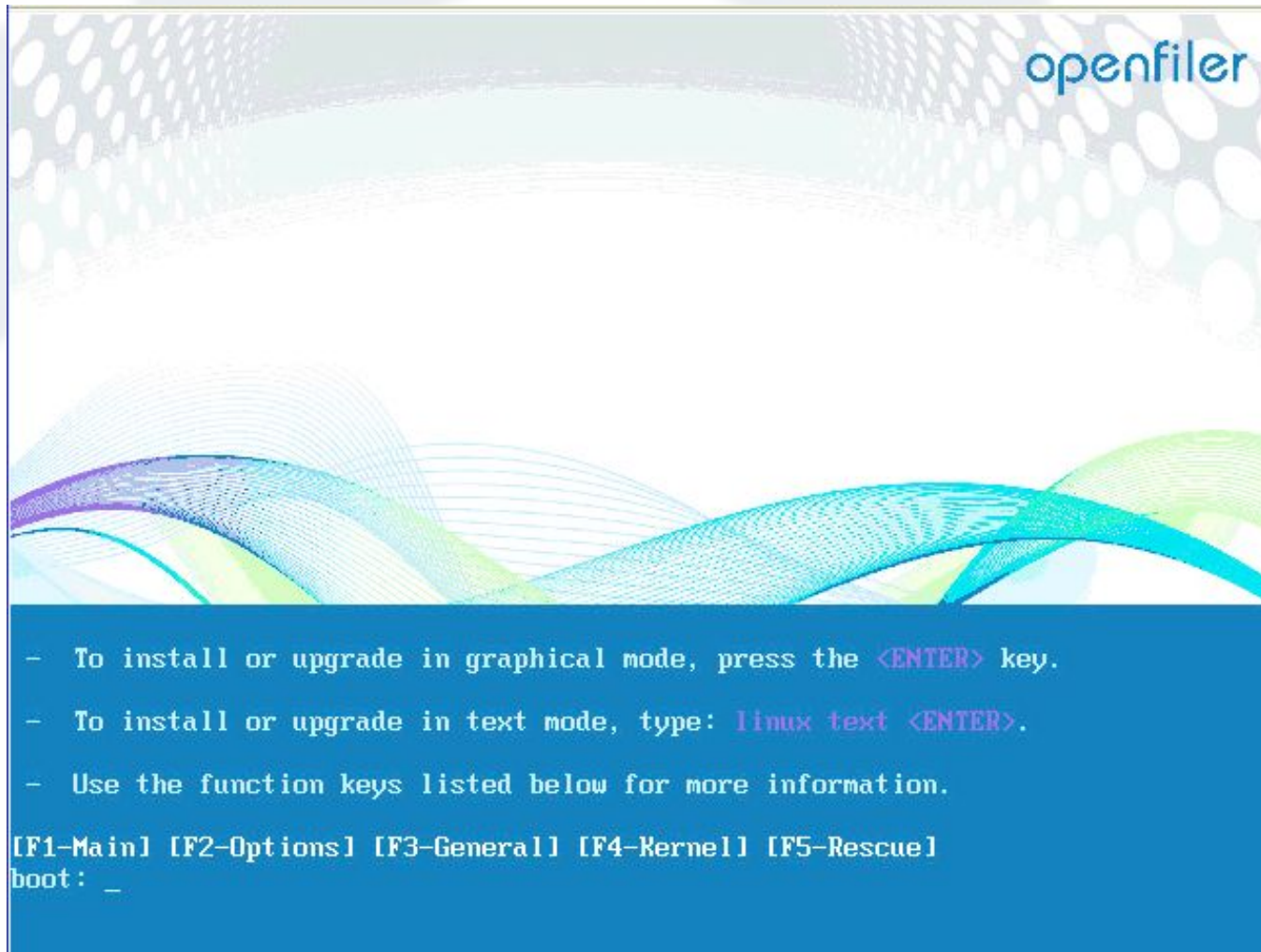
Boot from hard disk
Debian GNU/Linux      >
Ubuntu                 >
LiveOS                 >
Misc.                  >
*Special               >
About
```

```
                Miscellaneous Boot Options

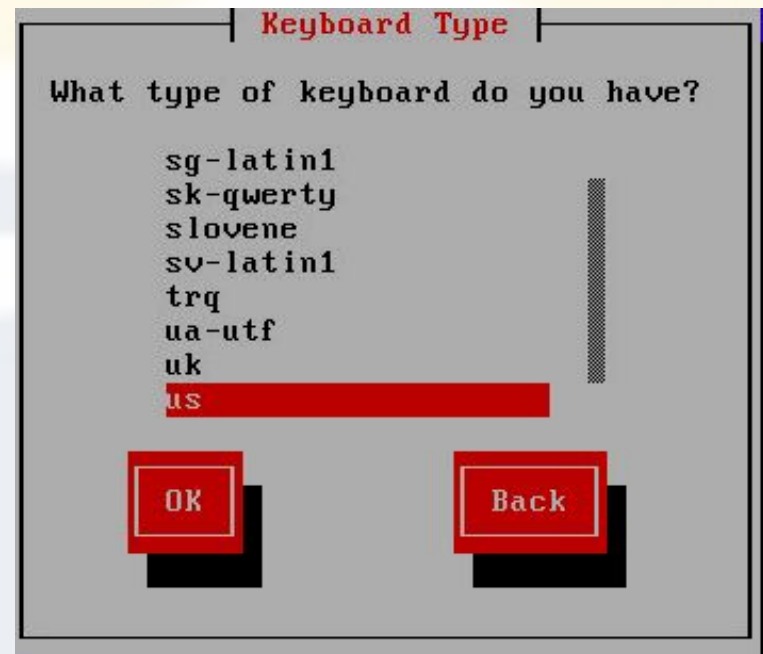
Back                   <
Run Memtest86+
Windows 10
*Run Memdisk/ISO loader (testing)
Security Onion (NSM)
*Openfiler 2.99
```

Tampilan awal dari instalasi

- Tekan enter

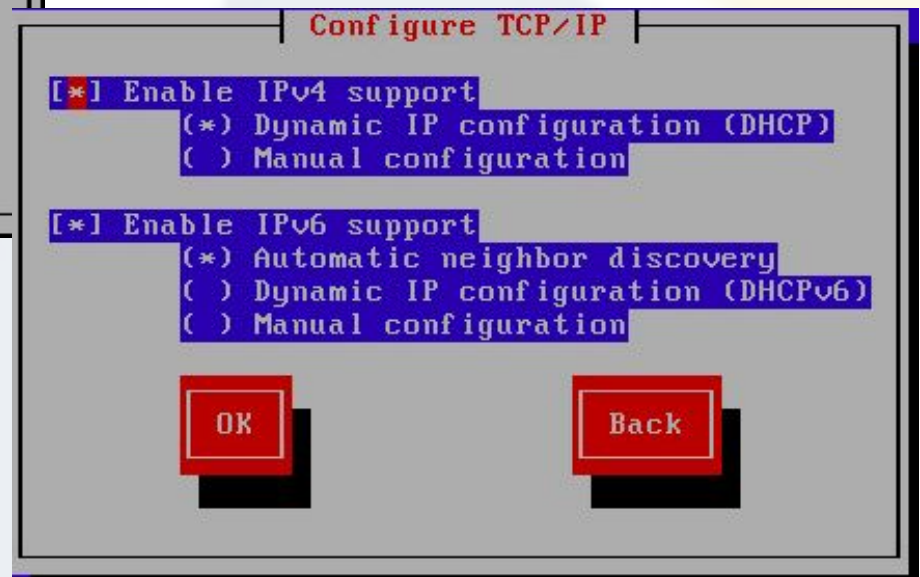


Pilih bahasa dan jenis keyboard



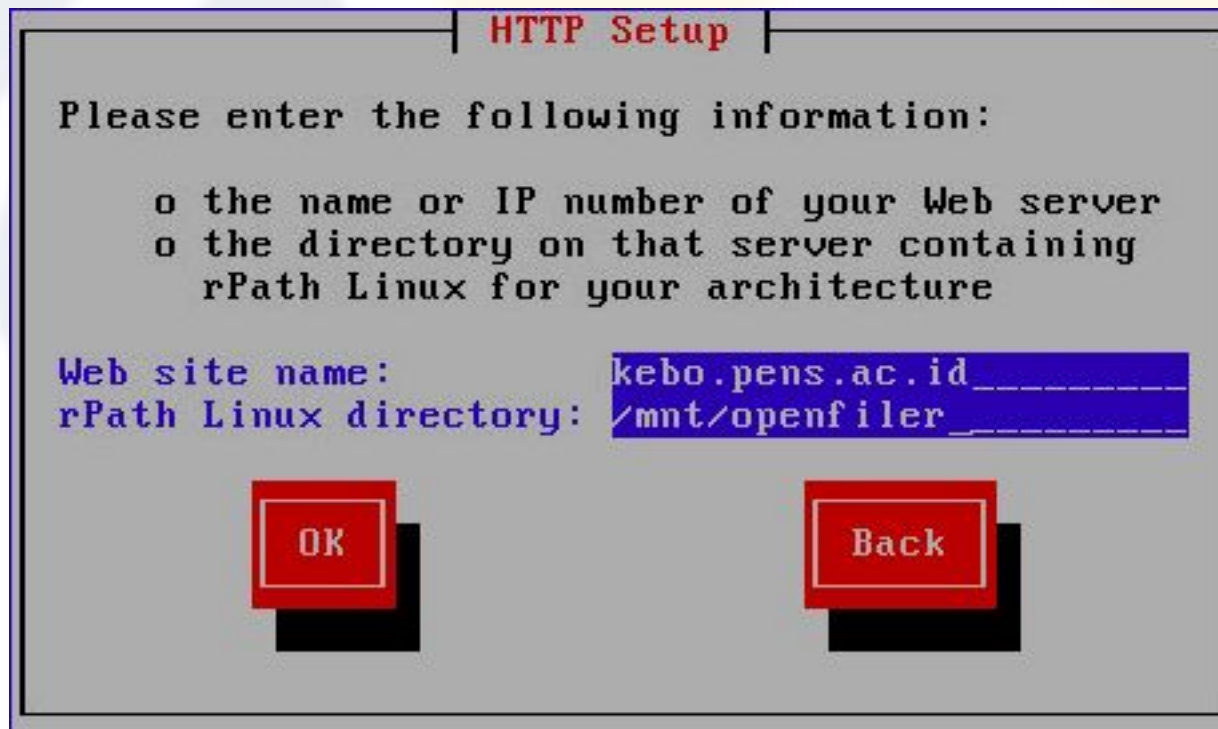
Pilih metode instalasi dan konfigurasi IP

- Pilih metode instalasi dengan HTTP



Konfigurasi instalasi HTTP

- Web site name: kebo.pens.ac.id
- rPath Linux directory: /mnt/openfiler



The image shows a screenshot of a 'HTTP Setup' dialog box. The title bar at the top is labeled 'HTTP Setup'. Inside the dialog, the text 'Please enter the following information:' is followed by two bullet points: 'o the name or IP number of your Web server' and 'o the directory on that server containing rPath Linux for your architecture'. Below these instructions, there are two input fields. The first is labeled 'Web site name:' and contains the text 'kebo.pens.ac.id'. The second is labeled 'rPath Linux directory:' and contains the text '/mnt/openfiler'. At the bottom of the dialog, there are two red buttons with white text: 'OK' on the left and 'Back' on the right.

HTTP Setup

Please enter the following information:

- o the name or IP number of your Web server
- o the directory on that server containing rPath Linux for your architecture

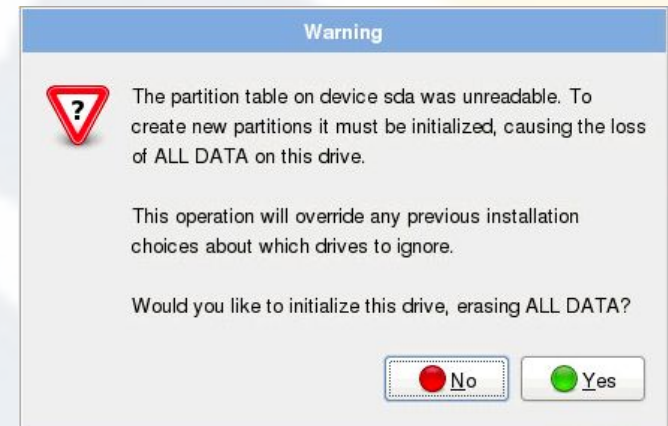
Web site name: kebo.pens.ac.id

rPath Linux directory: /mnt/openfiler

OK Back

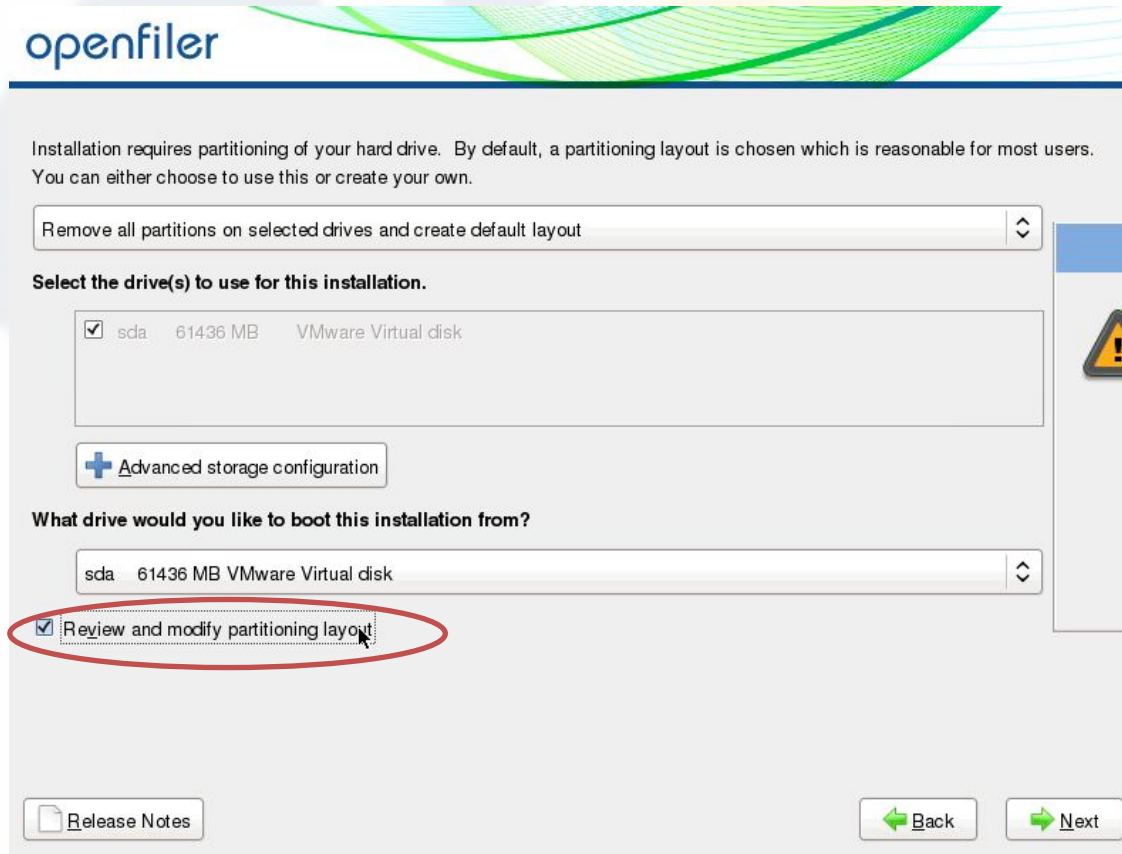
Instalasi dengan GUI siap dilanjutkan...

- Pastikan HD siap untuk diformat ulang



Konfigurasi HD

- Jangan lupa centang “review and modify partitioning layout



openfiler

Installation requires partitioning of your hard drive. By default, a partitioning layout is chosen which is reasonable for most users. You can either choose to use this or create your own.

Remove all partitions on selected drives and create default layout

Select the drive(s) to use for this installation.

☒ sda 61436 MB VMware Virtual disk

[+ Advanced storage configuration](#)

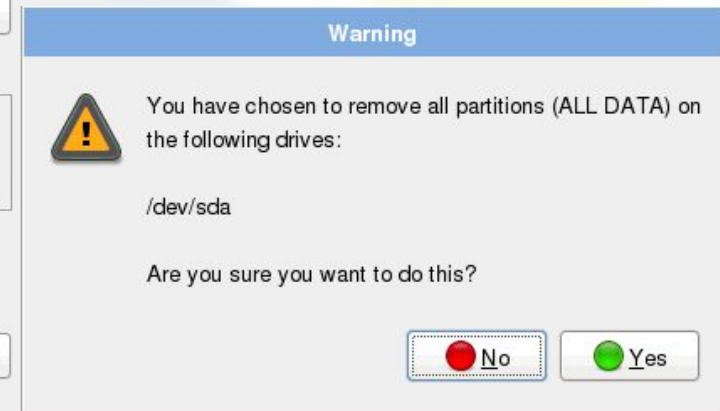
What drive would you like to boot this installation from?

sda 61436 MB VMware Virtual disk

☒ Review and modify partitioning layout

[Release Notes](#)

[Back](#) [Next](#)



Warning

 You have chosen to remove all partitions (ALL DATA) on the following drives:

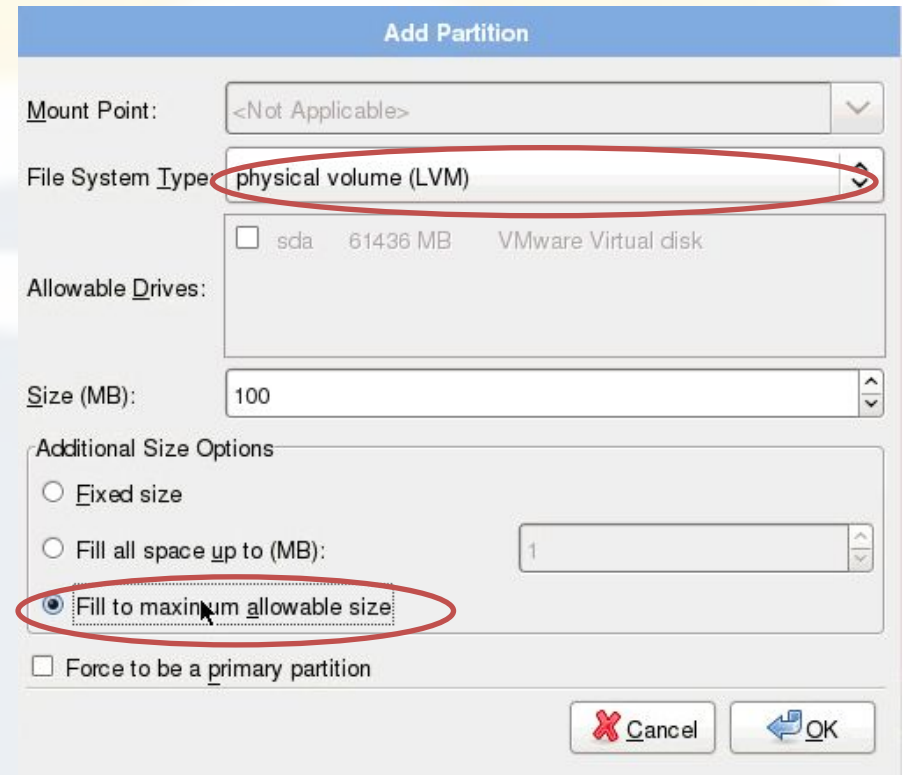
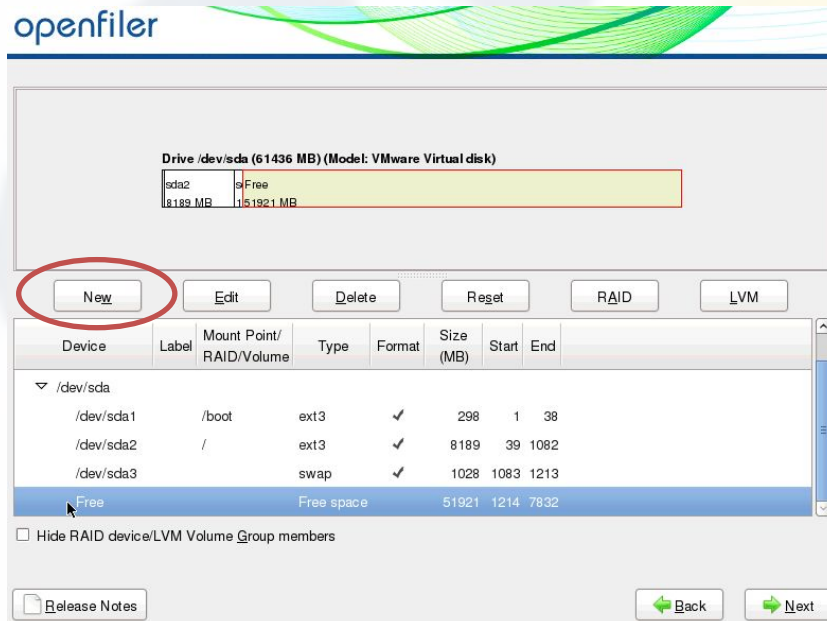
/dev/sda

Are you sure you want to do this?

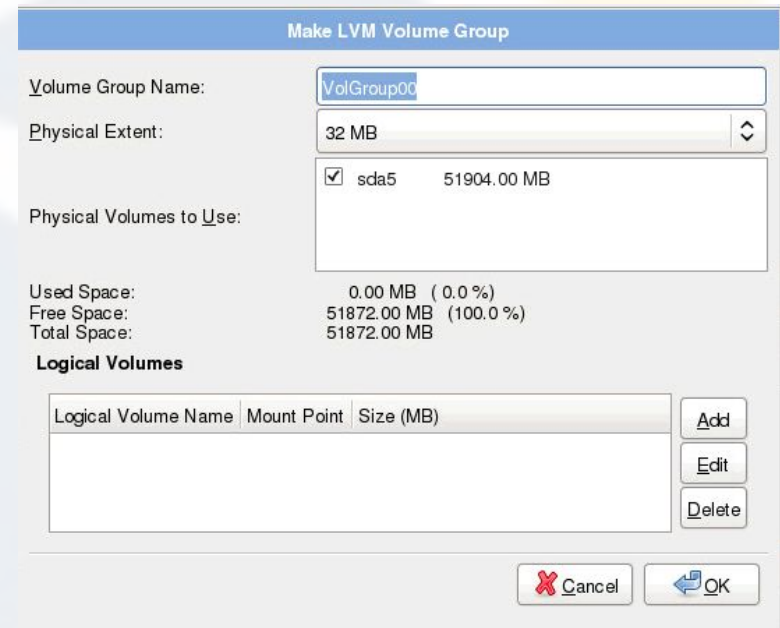
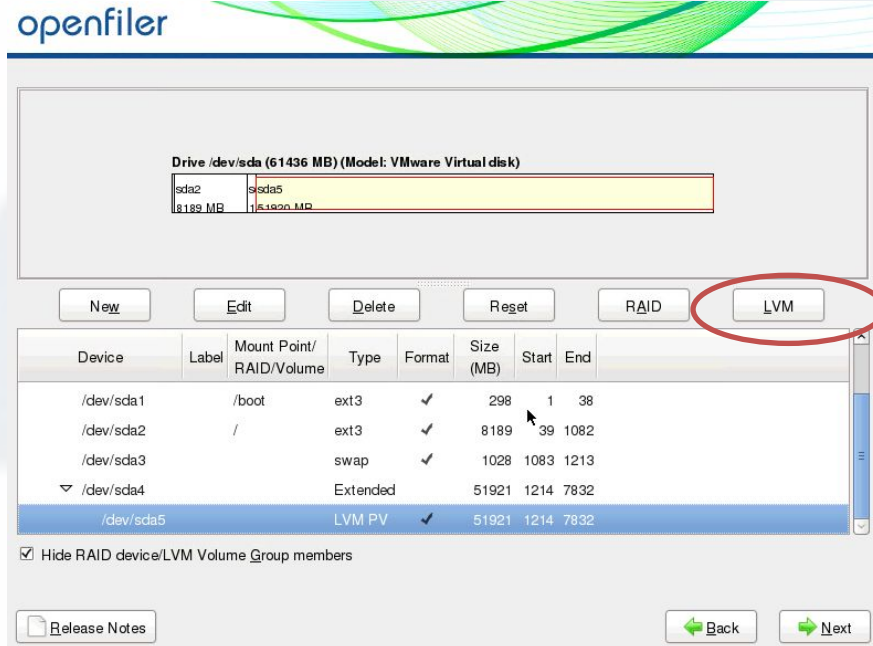
[No](#) [Yes](#)

Persiapkan partisi baru sebagai *storage*

- Klik “New” dan pilih LVM



Cek ulang partisi baru yang telah dibuat



Konfigurasi IP dan zona waktu

openfiler

Network Devices

Active on Boot	Device	IPv4/Netmask	IPv6/Prefix
☒	eth0	DHCP	Auto

Hostname

Set the hostname:

☒ automatically via DHCP

☐ manually (e.g., host.domain.com)

Miscellaneous Settings

Gateway:

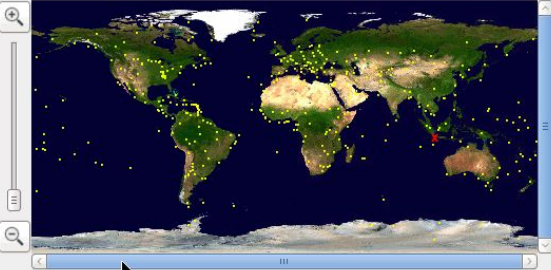
Primary DNS:

Secondary DNS:

[Release Notes](#) [Back](#) [Next](#)

openfiler

Please select the nearest city in your time zone:

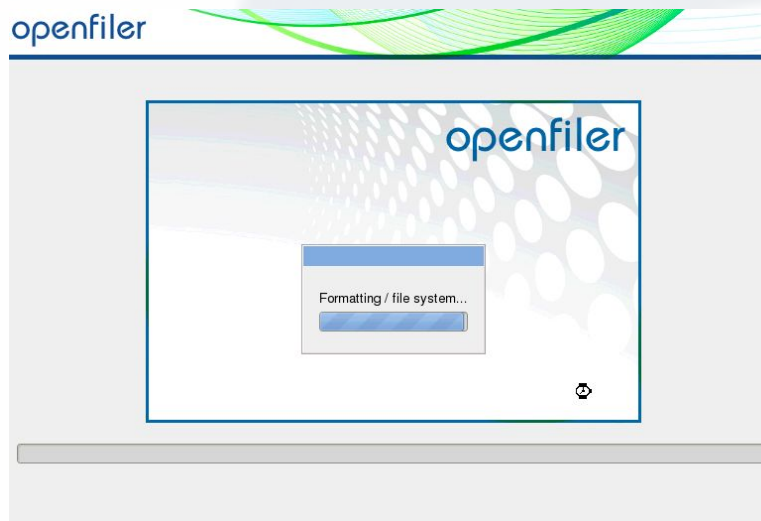
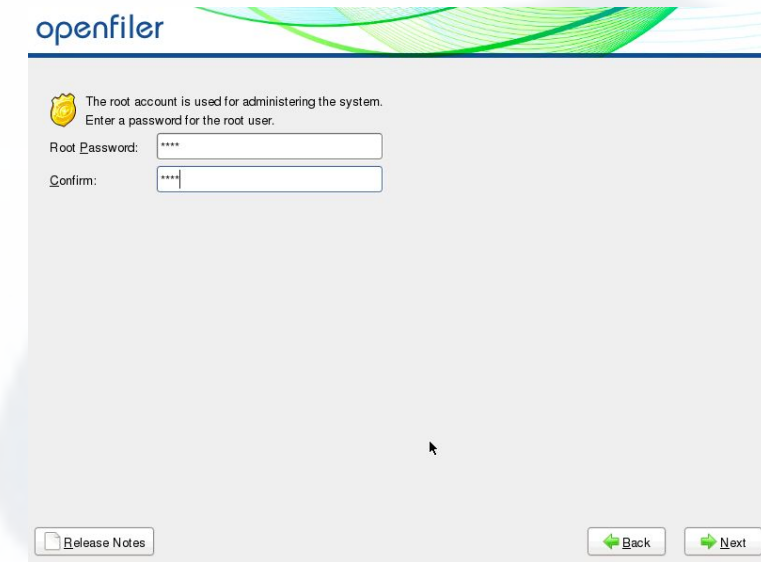


Selected city: Jakarta, Asia (Java & Sumatra)

☒ System clock uses UTC

[Release Notes](#) [Back](#) [Next](#)

Masukkan password root dan instalasi dimulai



Instalasi selesai

openfiler



Congratulations, the installation is complete.

Press the "Reboot" button to reboot your system.

 Release Notes

 Back

 Reboot

openf

OpenfilerESR(2.6.32-71.18.1.el6-0.20.smp.gcc4.1.x86_64)

Press ENTER to boot or TAB to edit a menu entry

Automatic boot in 2 seconds...

Hasil instalasi

```
  _/_/  _/_/_/  _/_/_/ _/  _/  _/  _/  _/  _/_/_/  _/
    _/
    _/

-----
:      Commercial Support: http://www.openfiler.com/support/      :
:  Administrator Guide: http://www.openfiler.com/buy/administrator-guide  :
:  Community Support: http://www.openfiler.com/community/forums/  :
:  Internet Relay Chat: server: irc.freenode.net    channel: #openfiler  :
-----
:      (C) 2001-2011 Openfiler. All Rights Reserved.      :
:  Openfiler is licensed under the terms of the GNU GPL, version 2  :
:      http://www.gnu.org/licenses/gpl-2.0.html  :
-----

Welcome to Openfiler ESA, version 2.99.1

Web administration GUI: https://10.252.209.193:446/

localhost login: _
```


Buka melalui browser di port 446



The image shows the Openfiler login interface. At the top, it says "openfiler" and "open source storage management". Below this, there are two input fields: "Username:" and "Password:". A "Log In" button is positioned below the password field. At the bottom left, it displays "Distro Release: Openfiler ESA 2.99.1" and "GUI Version: r_15ec1bb936e555e65adce24a9b0595124a46b0ee-1-1". A penguin icon is visible on the right side. At the bottom, there is a copyright notice: "© 2001 - 2011 Openfiler. All rights reserved." followed by links for "Home", "Documentation", "Support", "Website", and "License".

openfiler *open source storage management*

Username:

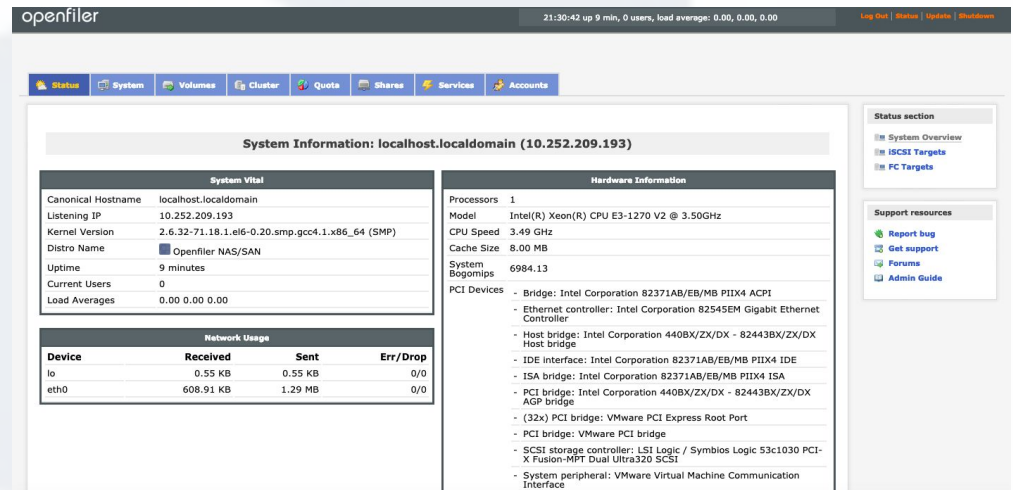
Password:

Log In

Distro Release: Openfiler ESA 2.99.1
GUI Version: r_15ec1bb936e555e65adce24a9b0595124a46b0ee-1-1

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[Home](#) • [Documentation](#) • [Support](#) • [Website](#) • [License](#)

- User: openfiler
- Pass: password



The image shows the Openfiler dashboard. At the top, it says "openfiler" and "21:30:42 up 9 min, 0 users, load average: 0.00, 0.00, 0.00". Below this, there is a navigation bar with tabs: "Status", "System", "Volumes", "Cluster", "Quota", "Shares", "Services", and "Accounts". The main content area is titled "System Information: localhost.localdomain (10.252.209.193)". It contains three tables: "System Vital", "Network Usage", and "Hardware Information".

System Information: localhost.localdomain (10.252.209.193)

System Vital			
Canonical Hostname	localhost.localdomain		
Listening IP	10.252.209.193		
Kernel Version	2.6.32-71.18.1.el6-0.20.smp.gcc4.1.x86_64 (SMP)		
Distro Name	Openfiler NAS/SAN		
Uptime	9 minutes		
Current Users	0		
Load Averages	0.00 0.00 0.00		

Network Usage			
Device	Received	Sent	Err/Drop
lo	0.55 KB	0.55 KB	0/0
eth0	608.91 KB	1.29 MB	0/0

Hardware Information	
Processors	1
Model	Intel(R) Xeon(R) CPU E3-1270 V2 @ 3.50GHz
CPU Speed	3.49 GHz
Cache Size	8.00 MB
System Bogomips	6984.13
PCI Devices	- Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI - Ethernet controller: Intel Corporation 82545EM Gigabit Ethernet Controller - Host bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX Host bridge - IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE - ISA bridge: Intel Corporation 82371AB/EB/MB PIIX4 ISA - PCI bridge: Intel Corporation 440BX/ZX/DX - 82443BX/ZX/DX AGP bridge - (32x) PCI bridge: VMware PCI Express Root Port - PCI bridge: VMware PCI bridge - SCSI storage controller: LSI Logic / Symbios Logic 53c1030 PCI-X Fusion-MPT Dual Ultra320 SCSI - System peripheral: VMware Virtual Machine Communication Interface

Status section:
[System Overview](#)
[iSCSI Targets](#)
[FC Targets](#)

Support resources:
[Report bug](#)
[Get support](#)
[Forums](#)
[Admin Guide](#)

Buka menu service

- Pastikan NFS “enable” dan ‘running’



Manage Services

Service	Boot Status	Modify Boot	Current Status	Start / Stop
CIFS Server	Disabled	Enable	Stopped	Start
NFS Server	Enabled	Disable	Running	Stop
RSync Server	Disabled	Enable	Stopped	Start
HTTP/Dav Server	Disabled	Enable	Running	Stop
LDAP Container	Disabled	Enable	Stopped	Start
FTP Server	Disabled	Enable	Stopped	Start
iSCSI Target	Disabled	Enable	Stopped	Start
UPS Manager	Disabled	Enable	Stopped	Start
UPS Monitor	Disabled	Enable	Stopped	Start
iSCSI Initiator	Disabled	Enable	Stopped	Start
ACPI Daemon	Enabled	Disable	Running	Stop
SCST Target	Disabled	Enable	Stopped	Start

Tambahkan volume baru

[Volumes](#) [Cluster](#) [Quota](#) [Shares](#) [Services](#) [Accounts](#)

Select Volume Group

Please select a volume group to display.

VolGroup00 [Change](#)

Volumes in volume group "VolGroup00" (51904 MB)



Create a volume in "VolGroup00"

Volume Name (*no spaces*. Valid characters [a-z,A-Z,0-9]):	disk1
Volume Description:	mydisk
Required Space (MB):	51904 
Filesystem / Volume type:	Ext3

[Create](#)

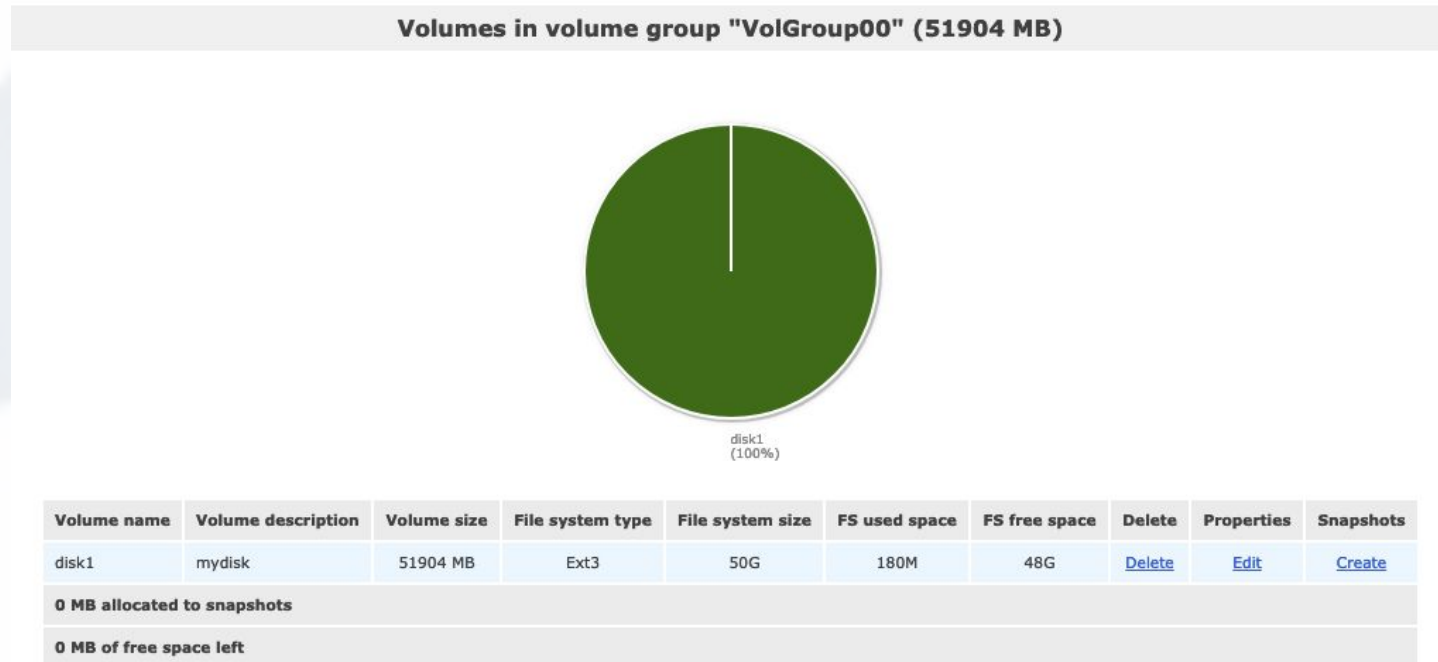
Volumes section

- [Manage Volumes](#)
- [Volume Groups](#)
- [Block Devices](#)
- [Add Volume](#)
- [iSCSI Targets](#)
- [Software RAID](#)

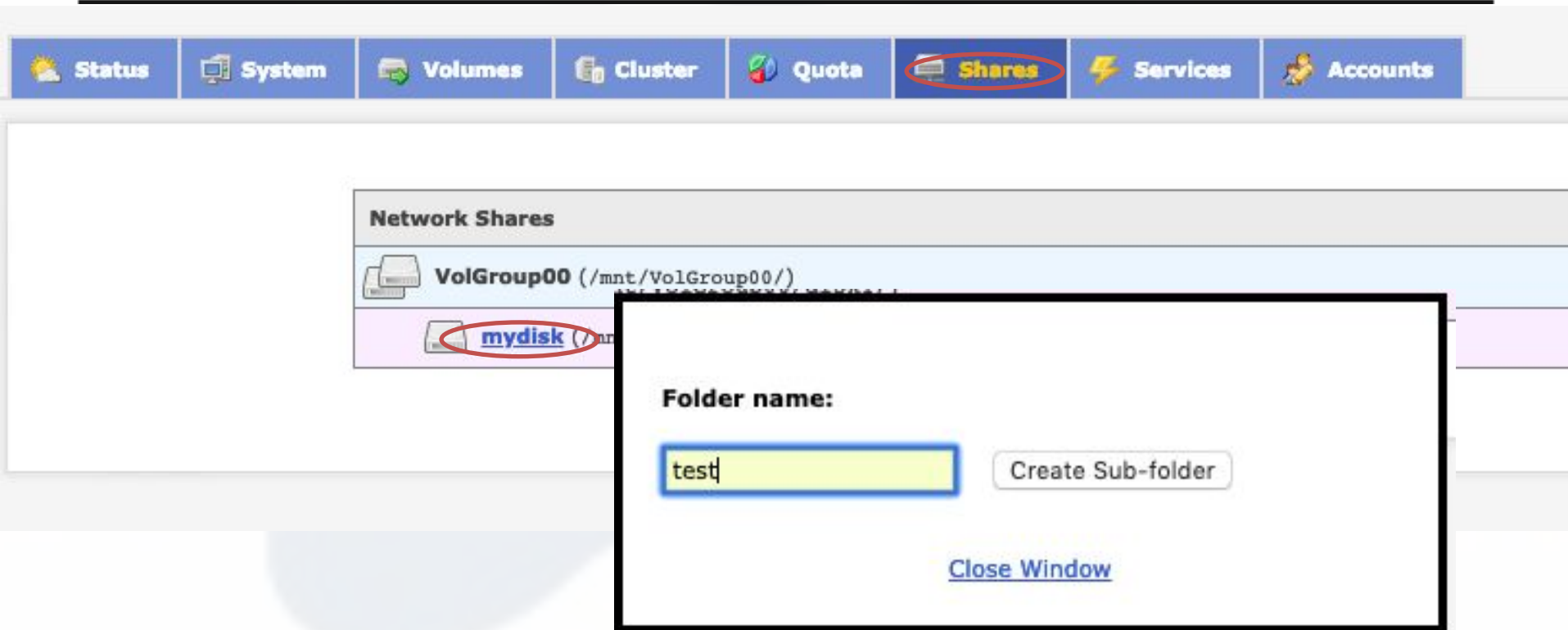
Support resources

- [Report bug](#)
- [Get support](#)
- [Forums](#)
- [Admin Guide](#)

Hasil tambah volume baru



Buat folder baru



Share direktori

[Status](#) [System](#) [Volumes](#) [Cluster](#) [Quota](#) [Shares](#) [Services](#) [Accounts](#)

Network Shares

	VolGroup00 (/mnt/VolGroup00/)
	mydisk (/mnt/VolGroup00/disk1/)
	test (/mnt/)

Shares section

-  [Existing Shares](#)
-  [Snapshot Shares](#)

Support resources

-  [Report bug](#)
-  [Get support](#)
-  [Forums](#)
-  [Admin Guide](#)

Folder name:

[Create Sub-folder](#)

New folder name:

[Rename Folder](#)

New description:

[Rename Description](#)

[Make Share](#) [Make Homes Share](#) [Delete Folder](#)

[Close Window](#)

Tambahkan IP atau network dari *client*

Folder name:

New folder name:

New description:

[Close Window](#)

Host access configuration (/mnt/VolGroup00/disk1/test/)

[\[Back to shares list \]](#)

A list of networks have not been created yet.
You cannot configure network access control unless
you create a list of networks in the [Local Networks](#) section.
Until that time, this share will be unavailable.

[\[Delete this share \]](#)

Interface	Boot Protocol	IP Address	Network Mask	Speed	MTU	Link	Edit
eth0	DHCP	10.252.209.193	255.255.255.0	1000Mb/s	1500	Yes	Configure

[Create bonded interface](#)

Network Access Configuration

Delete	Name	Network/Host	Netmask	Type
☐	jarkom	10.252.108.0	255.255.255.0	Share
New			0.0.0.0	Share

Mode kontrol akses berbagi

- Pada menu share folder/direktori

Share Access Control Mode

☒ **Public guest access**

☐ **Controlled access**

Update

Aktifkan layanan untuk masing IP atau jaringan

Host access configuration (/mnt/VolGroup00/disk1/test/)

[[Back to shares list](#)]

Name	Network	SMB/CIFS			NFS				HTTP(S) / WebDAV			FTP			Rsync		
		SMB/CIFS Options													Rsync Options		
		☒ Restart services															
		No	RO	RW	No	RO	RW	Options	No	RO	RW	No	RO	RW	No	RO	RW
jarkom	10.252.108.0	☒	☐	☐	☐	☐	☒	Edit	☒	☐	☐	☒	☐	☐	☒	☐	☐
galatea	202.9.85.208	☒	☐	☐	☐	☐	☒	Edit							☐	☐	

Update

[[Delete this share](#)]

NFS Share Client Options

(jarkom : 10.252.108.0)

Anonymous UID:	99
Anonymous GID:	99
UID/GID Mapping:	no_root_squash
I/O Mode:	sync
Write Delay:	wdelay
Request Origin Port:	secure (<1024)

Ujicoba *mount* di *client* (owncloud)

- # mkdir /mnt/owncloud
- # mount (ip addr dari openfiler):/mnt/VolGroup00/disk1/test /mnt/owncloud

```
root@debian:/home/student# df -h
Filesystem                               Size  Used Avail Use% Mounted on
/dev/sda1                                15G   1.6G   13G   11% /
udev                                     10M     0   10M    0% /dev
tmpfs                                    201M   4.5M  196M    3% /run
tmpfs                                    501M     0   501M    0% /dev/shm
tmpfs                                    5.0M     0   5.0M    0% /run/lock
tmpfs                                    501M     0   501M    0% /sys/fs/cgroup
10.252.209.193:/mnt/VolGroup00/disk1/test 50G  180M   48G    1% /mnt/owncloud
root@debian:/home/student#
```

Alt. NFS / SSHFS

Instalasi

- NFS

```
# apt install nfs-kernel-server
```

- SSHFS

```
# apt install sshfs ssh
```

Konfigurasi (mount) NFS

- *Export NFS*

```
# vim /etc/exports
```

```
/nfs 10.252.13.91(rw,sync,no_subtree_check,no_root_squash)
```

```
# systemctl restart nfs-kernel-server
```

Konfigurasi (mount) NFS/SSHFS

- *mount* secara manual

```
# mount -t nfs 10.252.13.90:/mnt /mnt/nfs
```

```
# sshfs idris@10.252.13.90:/mnt /mnt/ssh
```

- *mount* secara otomatis

```
# vim /etc/fstab
```

```
<file system> <mount point> <type> <options> <dump> <pass>
```

```
10.252.13.90:/mnt /mnt/nfs nfs defaults,mountvers=3,nfsvers=3 0 0
```

Jika pakai LXC

```
# mknod -m 666 /dev/fuse c 10 229
```


Instalasi ownCloud

Update repository

```
root@debian:/home/student# apt-get update
Ign http://kebo.pens.ac.id jessie InRelease
Hit http://kebo.pens.ac.id jessie Release.gpg
Hit http://kebo.pens.ac.id jessie Release
Hit http://kebo.pens.ac.id jessie/main Sources
Hit http://kebo.pens.ac.id jessie/non-free Sources
Hit http://kebo.pens.ac.id jessie/contrib Sources
Hit http://kebo.pens.ac.id jessie/main amd64 Packages
Hit http://kebo.pens.ac.id jessie/non-free amd64 Packages
Hit http://kebo.pens.ac.id jessie/contrib amd64 Packages
Hit http://kebo.pens.ac.id jessie/contrib Translation-en
Hit http://kebo.pens.ac.id jessie/main Translation-en
Hit http://kebo.pens.ac.id jessie/non-free Translation-en
Reading package lists... Done
root@debian:/home/student#
```

Mount NFS dari NAS

```
root@debian:/home/student# mkdir /mnt/owncloud
root@debian:/home/student# mount 10.252.209.193:/mnt/VolGroup00/disk1/test /mnt/owncloud/
root@debian:/home/student# df -h
```

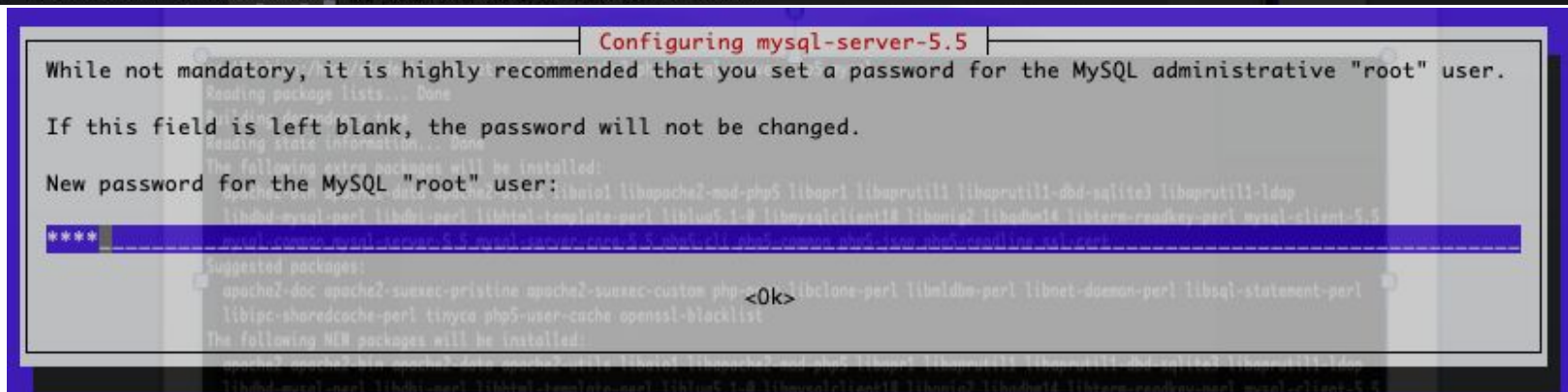
Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	15G	1.6G	13G	11%	/
udev	10M	0	10M	0%	/dev
tmpfs	201M	4.5M	196M	3%	/run
tmpfs	501M	0	501M	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	501M	0	501M	0%	/sys/fs/cgroup
10.252.209.193:/mnt/VolGroup00/disk1/test	50G	180M	48G	1%	/mnt/owncloud

```
root@debian:/home/student#
```

Instalasi aplikasi pendukung

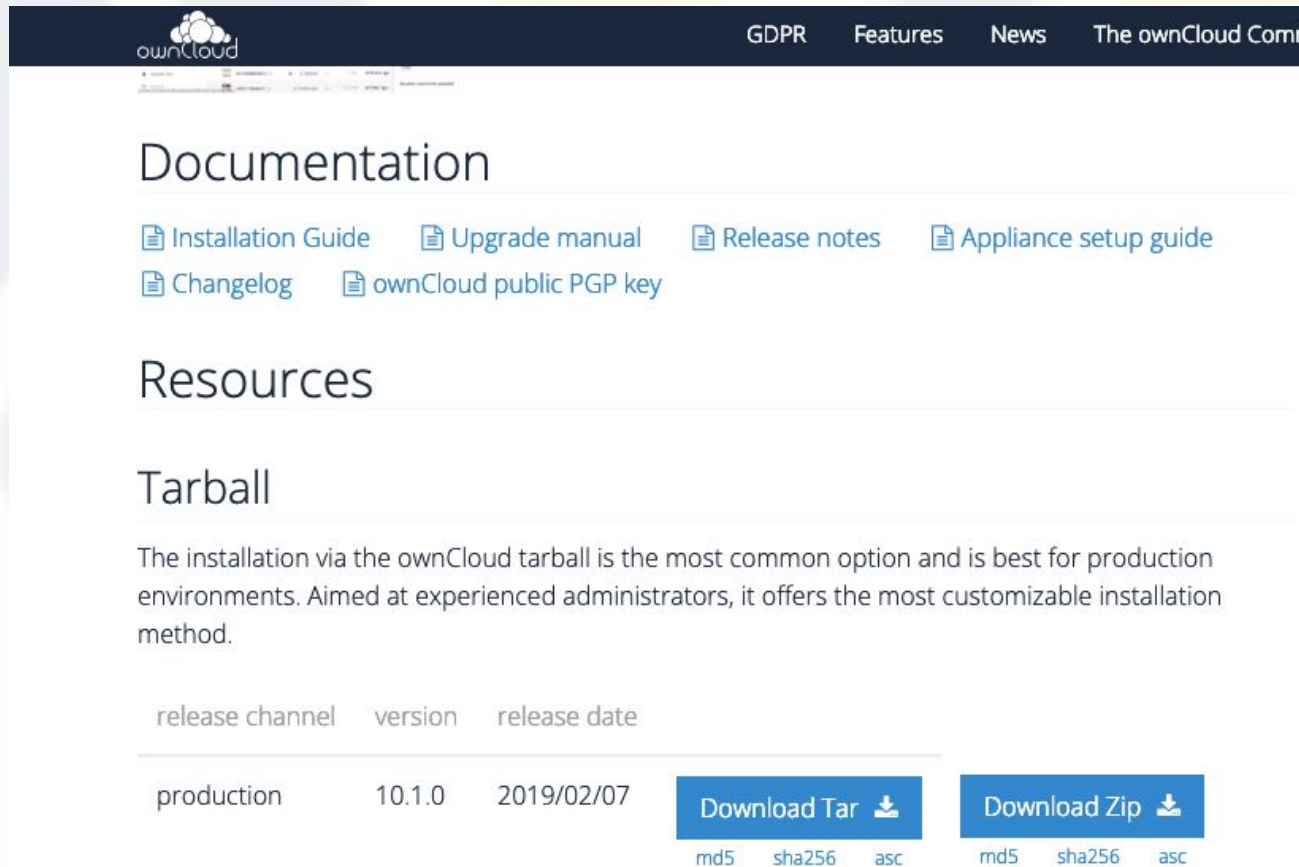
apt-get install apache2 mysql-server php5-json php5-curl php5-mcrypt
php5-mysqldb php5-pgsql php5-gd php5-imagick php5-intl phpmyadmin

```
root@debian:/var/www/html/owncloud# apt-get install apache2 mysql-server php5-json php5-curl php5-mcrypt php5-mysqldb php5-pgsql php5-gd php5-imagick php5-intl
Reading package lists... Done
Building dependency tree
Reading state information... Done
mysql-server is already the newest version.
php5-json is already the newest version.
php5-mcrypt set to manually installed.
The following extra packages will be installed:
  fonts-droid ghostscript gsfonts imagemagick-common libcupsfilters1 libcupsimage2 libcurl3 libfftw3-double3 libgd3 libgs9 libgs9-common
  libijs-0.35 libjbig2dec0 liblqr-1-0 libltdl7 libmagickcore-6.q16-2 libmagickwand-6.q16-2 libmcrypt4 libpq5 libvpx1 libxpm4 poppler-data
  ttf-dejavu-core
Suggested packages:
  ghostscript-x libfftw3-bin libfftw3-dev libgd-tools libmagickcore-6.q16-2-extra libmcrypt-dev mcrypt poppler-utils fonts-japanese-mincho
  fonts-ipafont-mincho fonts-japanese-gothic fonts-ipafont-gothic fonts-arphic-ukai fonts-arphic-uming fonts-nanum
The following packages will be REMOVED:
  php5-mysql
The following NEW packages will be installed:
  fonts-droid ghostscript gsfonts imagemagick-common libcupsfilters1 libcupsimage2 libcurl3 libfftw3-double3 libgd3 libgs9 libgs9-common
  libijs-0.35 libjbig2dec0 liblqr-1-0 libltdl7 libmagickcore-6.q16-2 libmagickwand-6.q16-2 libmcrypt4 libpq5 libvpx1 libxpm4 php5-curl
  php5-gd php5-imagick php5-intl php5-mcrypt php5-mysqldb php5-pgsql poppler-data ttf-dejavu-core
0 upgraded, 30 newly installed, 1 to remove and 0 not upgraded.
Need to get 17.7 MB of archives.
After this operation, 64.3 MB of additional disk space will be used.
```



Download ownCloud

- http://idris.lecturer.pens.ac.id/files/admin_jarkom/owncloud-10.1.0.tar.bz2



The screenshot shows the ownCloud website's download page. The header includes the ownCloud logo and navigation links for GDPR, Features, News, and The ownCloud Community. The main content area is titled 'Documentation' and lists links for Installation Guide, Upgrade manual, Release notes, Appliance setup guide, Changelog, and ownCloud public PGP key. Below this is a 'Resources' section, followed by a 'Tarball' section. The 'Tarball' section explains that the tarball is the most common option for production environments. At the bottom, there is a table with columns for release channel, version, and release date. The table shows the 'production' channel, version '10.1.0', and release date '2019/02/07'. To the right of the table are two buttons: 'Download Tar' and 'Download Zip', each with a download icon and links for md5, sha256, and asc checksums.

release channel	version	release date		
production	10.1.0	2019/02/07	Download Tar 	Download Zip 
			md5 sha256 asc	md5 sha256 asc

Download dan ekstrak owncloud

```
root@debian:/var/www/html# wget http://idris.lecturer.pens.ac.id/files/admin_jarkom/owncloud-10.1.0.tar.bz2
converted 'http://idris.lecturer.pens.ac.id/files/admin_jarkom/owncloud-10.1.0.tar.bz2' (ANSI_X3.4-1968) -> 'http://idris.lecturer.pens.ac.id/files/admin_jarkom/owncloud-10.1.0.tar.bz2' (UTF-8)
--2019-03-06 07:51:05-- http://idris.lecturer.pens.ac.id/files/admin_jarkom/owncloud-10.1.0.tar.bz2
Resolving idris.lecturer.pens.ac.id (idris.lecturer.pens.ac.id)... 202.9.85.69
Connecting to idris.lecturer.pens.ac.id (idris.lecturer.pens.ac.id)|202.9.85.69|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20345340 (19M) [application/x-bzip2]
Saving to: 'owncloud-10.1.0.tar.bz2'

owncloud-10.1.0.tar.bz2      100%[=====>] 19.40M  111MB/s  in 0.2s
                                3 members found this
2019-03-06 07:51:05 (111 MB/s) - 'owncloud-10.1.0.tar.bz2' saved [20345340/20345340]
```

```
root@debian:/var/www/html# tar jxvf owncloud-10.1.0.tar.bz2
owncloud/
owncloud/AUTHORS
owncloud/occ
owncloud/config/
owncloud/config/config.apps.sample.php
owncloud/config/config.sample.php
owncloud/cron.php
owncloud/robots.txt
owncloud/.user.ini
owncloud/console.php
owncloud/ocs/
```

Buat user dan db baru via phpmyadmin



Welcome to phpMyAdmin

Language

English

Log in

Username: root

Password:

Go

Server: localhost

Databases SQL Status Users Export Import Settings Re

Add user

Login Information

User name: Use text field: owncloud

Host: Any host %

Password: Use text field: ..

Re-type: ..

Generate password: Generate

Database for user

- ☒ Create database with same name and grant all privileges.
- ☒ Grant all privileges on wildcard name (username_%).

Buka instalasi via browser

```
[root@debian:/mnt# mkdir /mnt/owncloud  
[root@debian:/mnt# chmod 777 /mnt/owncloud/  
root@debian:/mnt#
```



Create an admin account

admin

admin

Very weak password

Storage & database ▾

Data folder

/mnt/owncloud

Configure the database

MySQL/MariaDB PostgreSQL

owncloud

..

owncloud

localhost

Please specify the port number along with the host name (e.g., localhost: 5432).

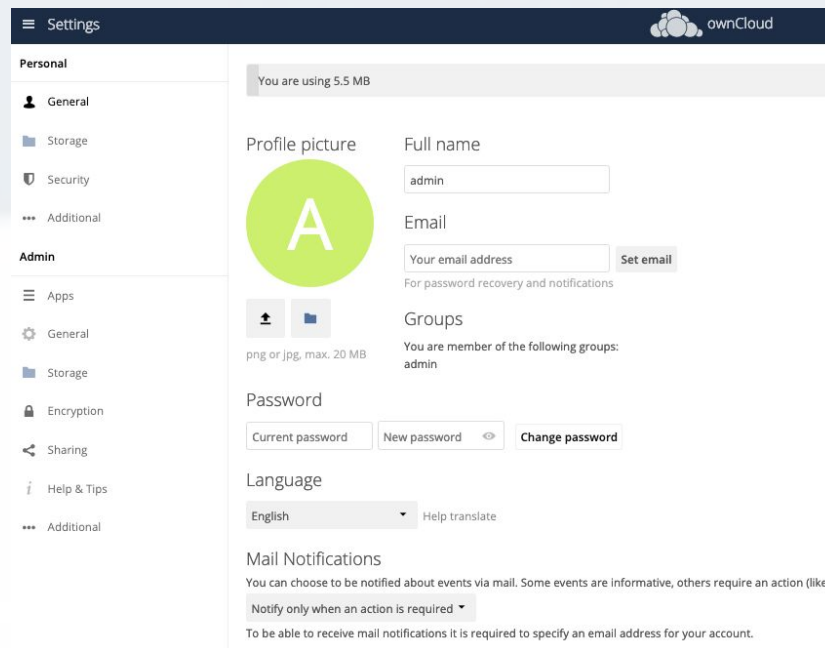
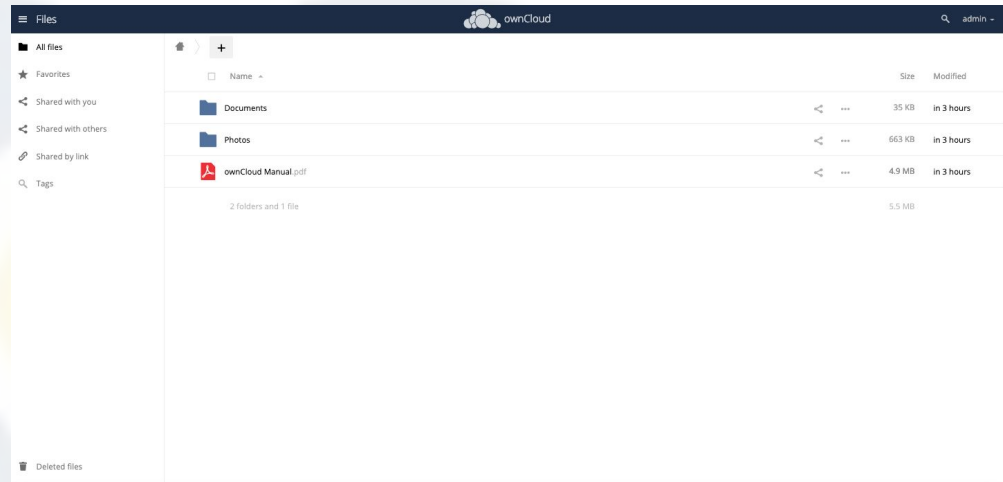
Modifikasi direktori data

Your Data directory is readable by other users

Please change the permissions to 0770 so that the directory cannot be listed by other users.

```
root@debian:~# cd /mnt/andri
root@debian:/mnt# chmod 0770 owncloud/
root@debian:/mnt# chown www-data:www-data owncloud/
root@debian:/mnt#
```

Owncloud siap digunakan



Ujicoba di *mobile*

- Silahkan download owncloud di App Store (iPhone) atau di playstore (Android)